F2G SOLUTIONS INC × KFOKAM48 TELECOMMUNICATION ACADEMY

CMAS / PWS
For Cameroon

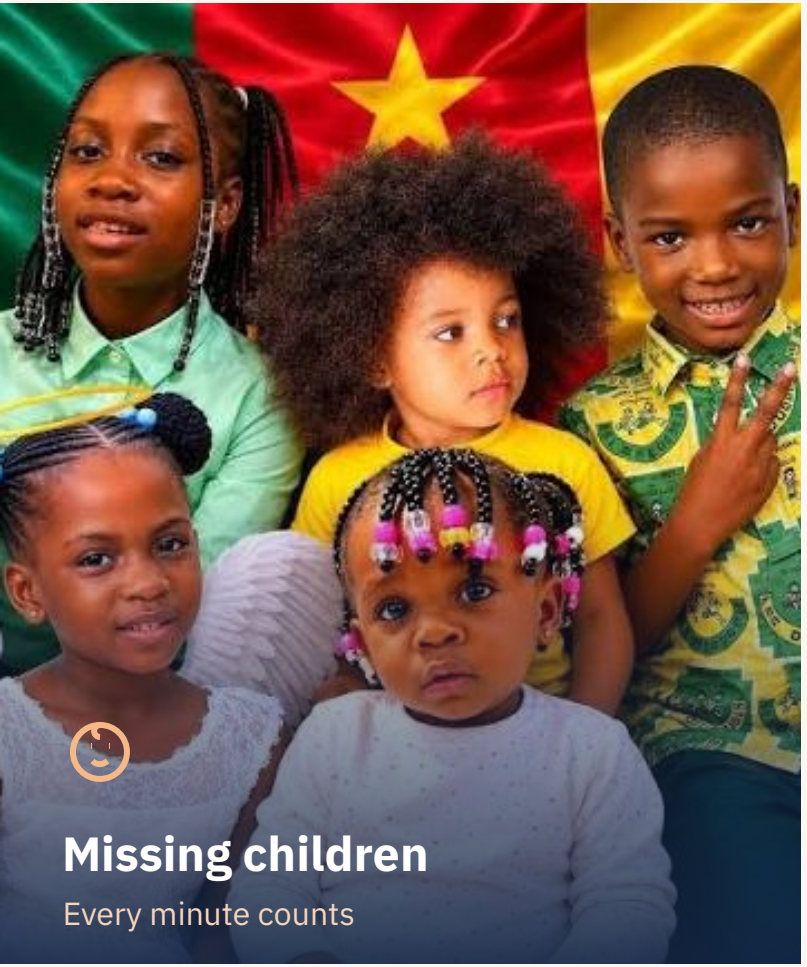
One alert. Every phone.
Under four seconds.

A national emergency cell-broadcast alert system



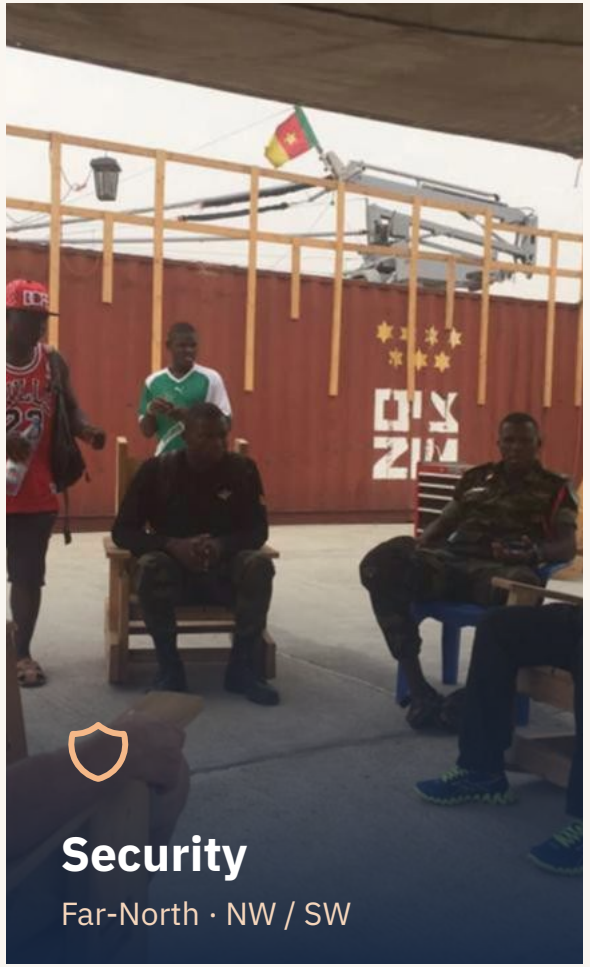
CONTEXT

Cameroon's emergencies keep coming back



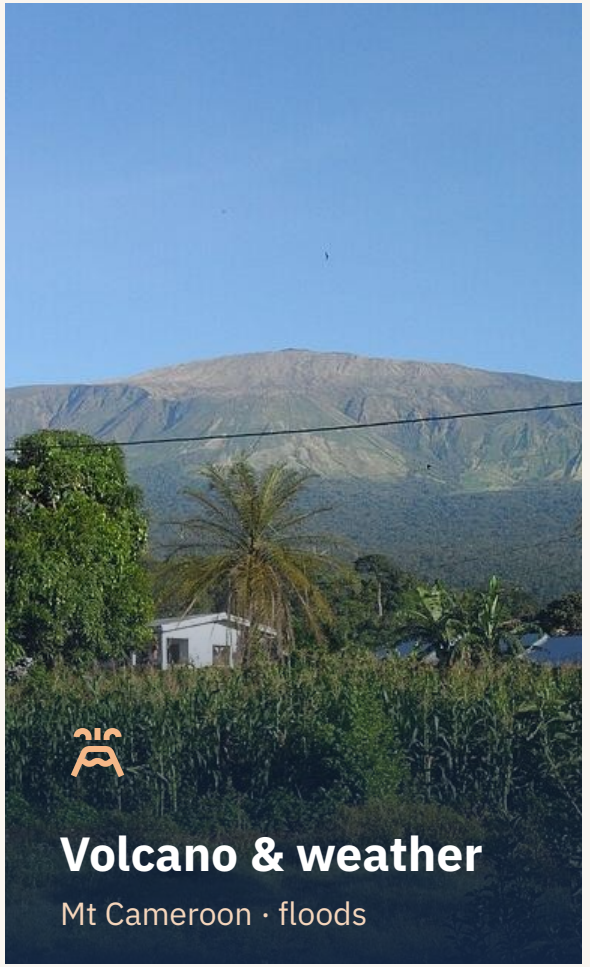
Missing children

Every minute counts



Security

Far-North · NW / SW



Volcano & weather

Mt Cameroon · floods



Public health

Cholera · meningitis

THE PROBLEM

When disaster strikes, Cameroon cannot reach every phone in the danger zone within seconds.

Mass SMS, radio and social media take minutes to hours.



THE SOLUTION · F2G CMAS HUB

CM-Alert warns a whole area in under 4 seconds

One platform. Every operator. Any target area.

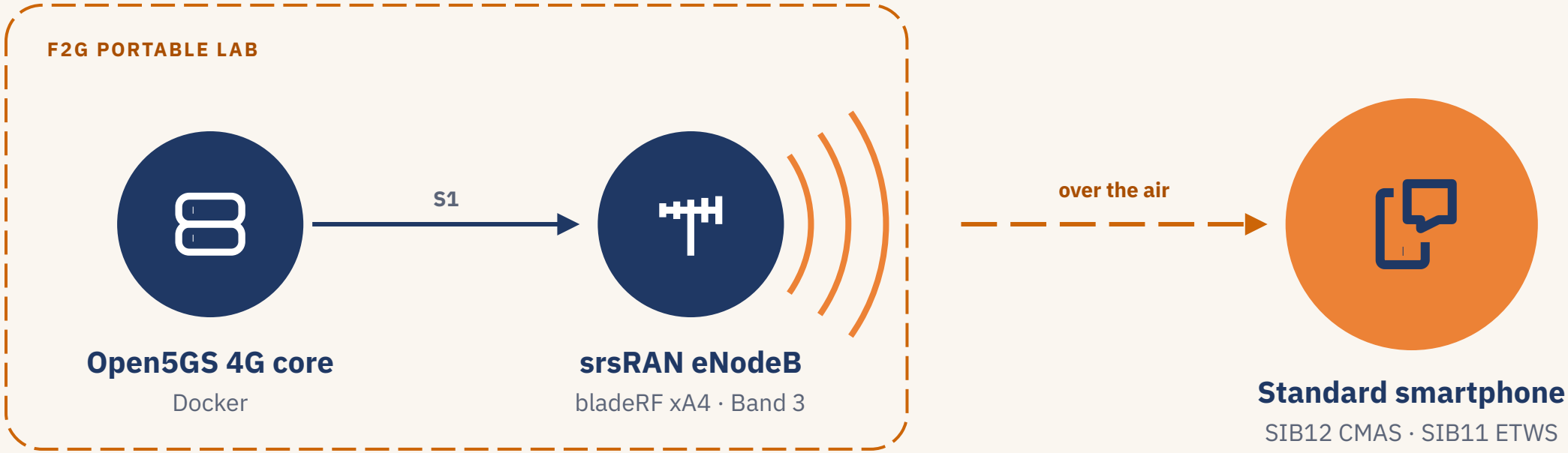
Fast
Under 4 seconds to every handset in range

No numbers needed
Targets cells, not SIM lists

Unmissable
Full-screen popup with a loud alert tone

PROTOTYPE

Already working in our lab



10 **alert scenarios tested**

All captured in Wireshark

NEXT IN THE LAB

TAC geo-targeting

Yaoundé TAC 1 vs Douala TAC 2

TEAM

Field RAN engineering meets KFOKAM48 / PKF

AD

F2G SOLUTIONS INC

Arnaud DJOUM

Telecom engineer

- RAN I&C for Nokia / Safaricom Ethiopia, 2026
- Private 4G/5G lab
- Built the CMAS prototype

×

LG

KFOKAM48 / PKF

Luis Geralda

KFOKAM48 Telecommunication Academy

VISION

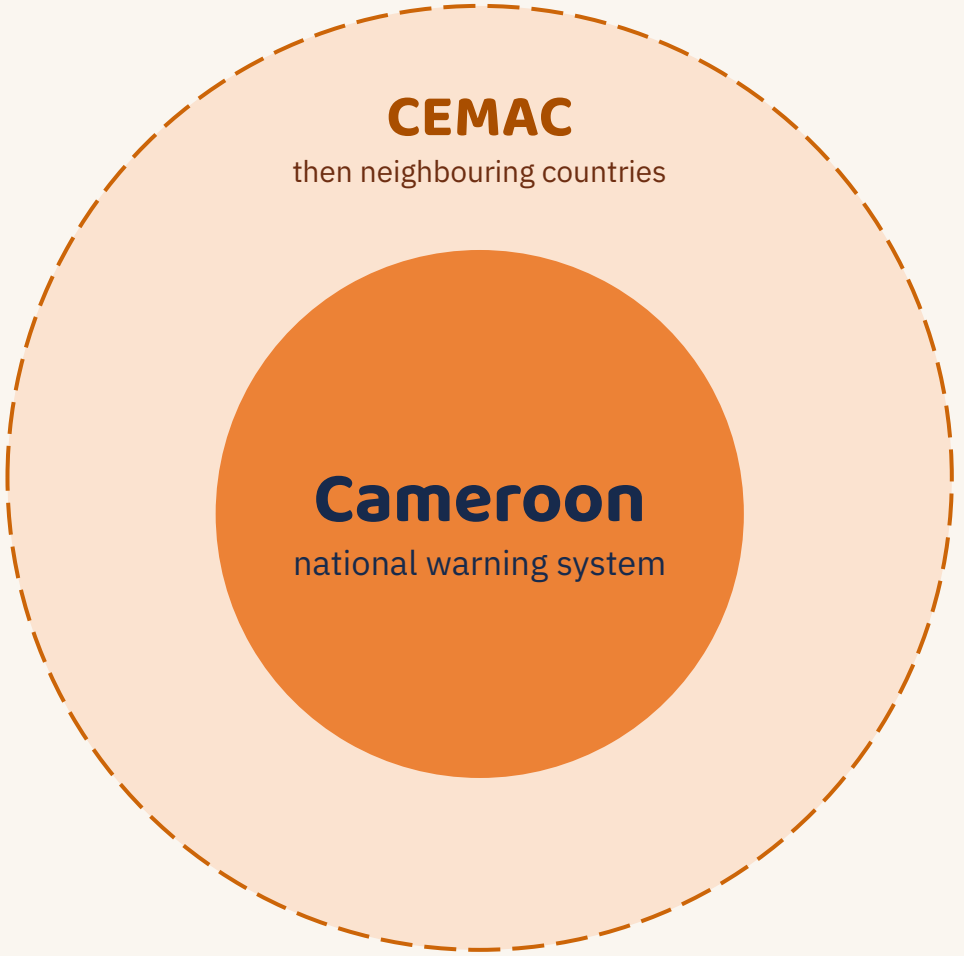
CM-Alert: Cameroon first, then all of CEMAC

USE CASES

- AMBER alerts for missing children
- Mt Cameroon, Far-North floods
- Security incidents
- Health emergencies

TARGET CUSTOMERS

MINAT · Protection Civile | **MINSANTE** | **MINDEF**

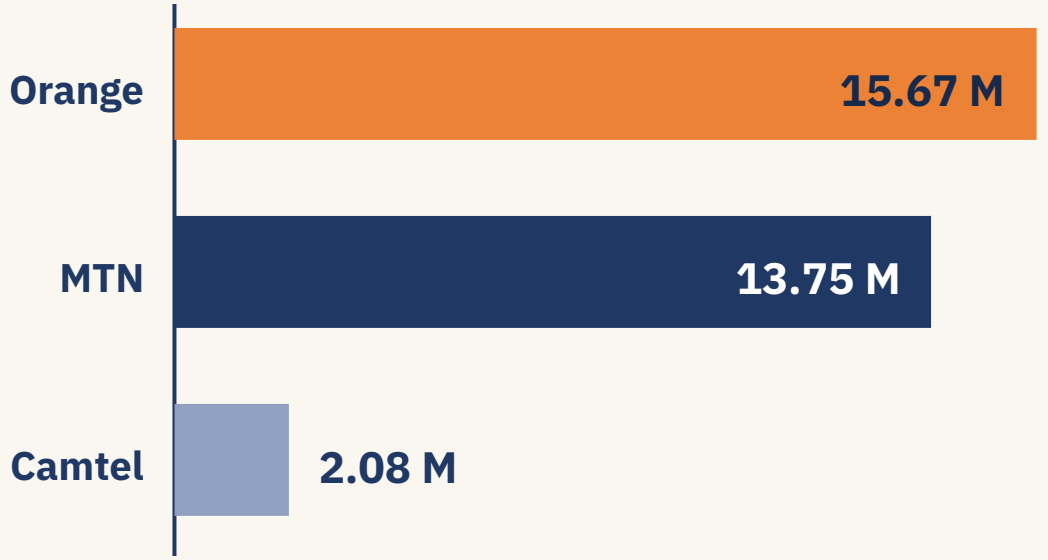


31.5 M mobile subscriptions to protect

31.5 M

active mobile subscriptions

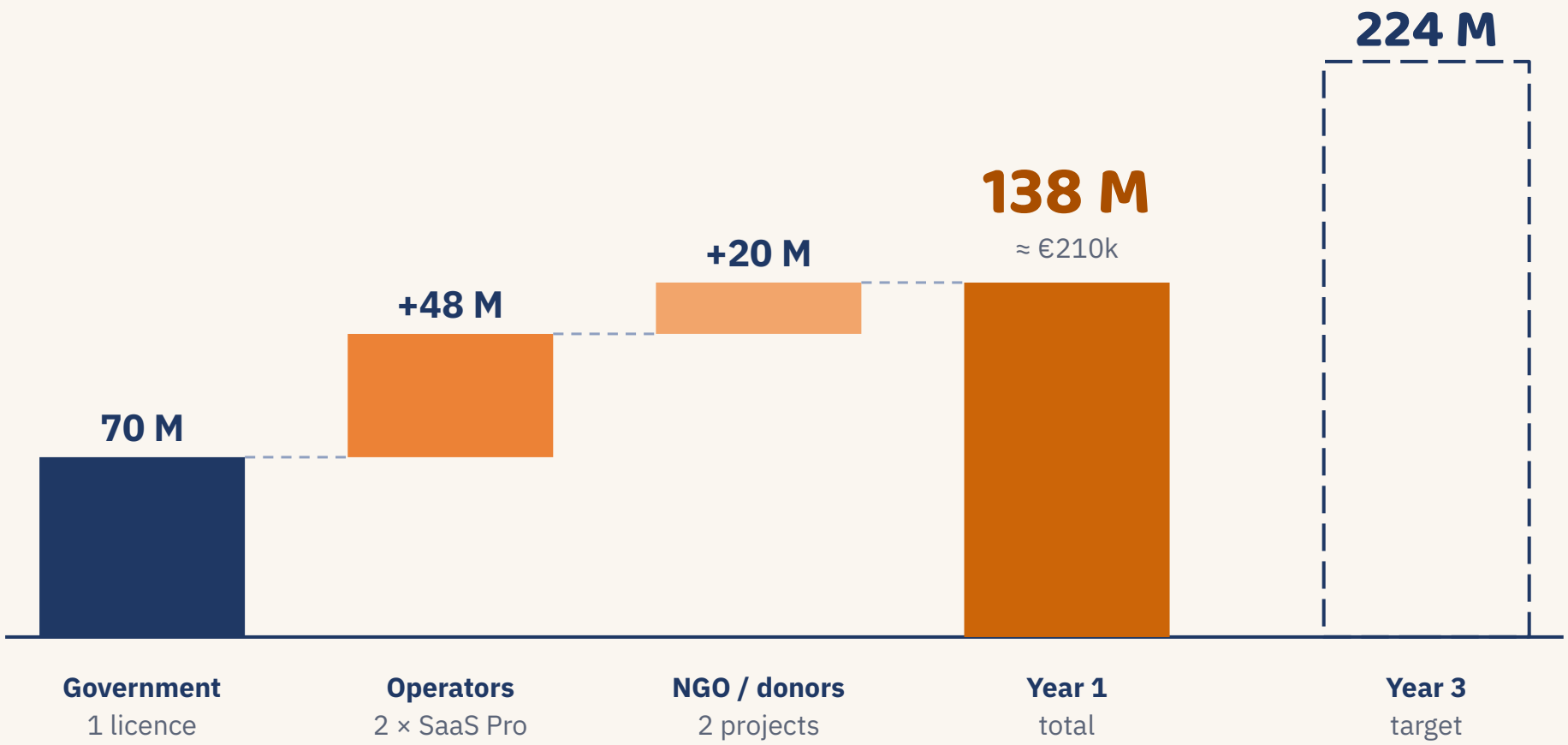
Cameroon, 2024



Source: ART Cameroon, 2024 data (TechAfrica News, Dec 2025).
Nexttel, not in this count, reports about 5 M users.

Year 1: 138 M FCFA from three buyer groups

Revenue, M FCFA



F2G ESTIMATE

Bottom-up, from our price list

Regulator first — then every operator needs us

> **Once the regulator mandates CMAS, every operator needs a broadcast centre.**



MINPOSTEL · ART
National framework

MTN or Orange
LOI, SBc-AP link, one city

MINAT · MINSANTE MINDEF

All 4 operators
then CEMAC

TRACTION

No customer yet, but a working prototype

DONE

Working lab prototype
10 alert scenarios

21 Sep 2026
Internal pitch, PKF Telco Academy

EN / FR technical guide
and demo scripts

NOW

NEXT PROOF POINTS

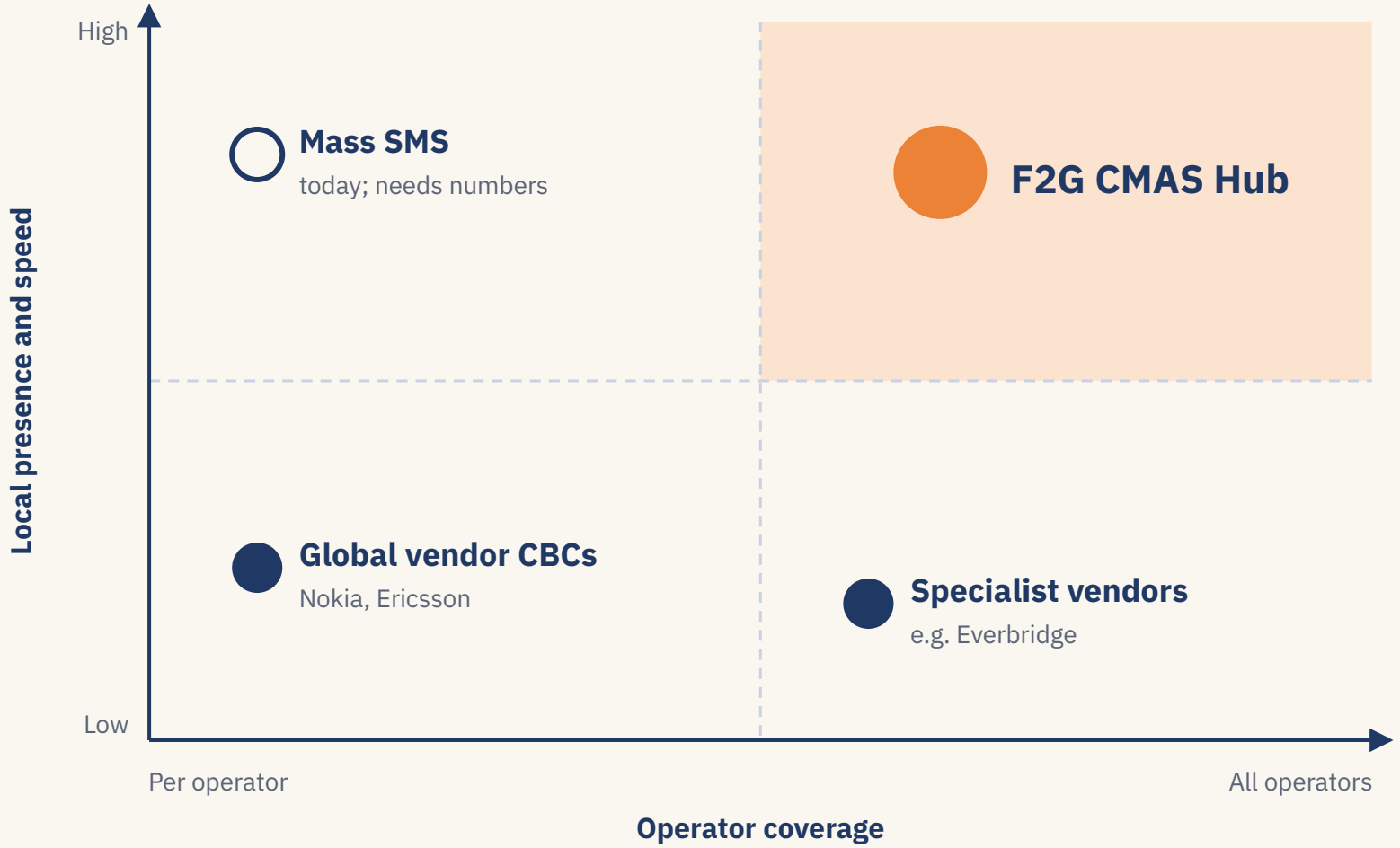
MINPOSTEL meeting

One operator LOI

TAC-targeted demo

COMPETITION

F2G is the local, all-operator option



WHY F2G SITS THERE

- One hub, every operator
- Open 3GPP standards
- Local, bilingual team
- Working prototype

F2G reading,
not market data.

WHY WE WIN

Same 3GPP standard, delivered locally

	F2G CMAS Hub	Global vendors
Local engineers	**In Cameroon**	Support from abroad
Bilingual support	**English and French**	English-first
Operator-neutral	**One hub, all operators**	Deployed per operator
Proof in Cameroon	**Lab prototype today**	Proven abroad, heavy budget

TIMELINE

From lab to national service in 21 months

POC · 3 months · 15 M FCFA

Core engine and dashboard MVP

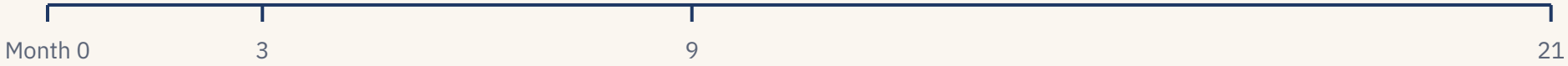
Pilot · 6 months · 50 M FCFA

MINPOSTEL plus one operator, one city

National · 12 months · 100 M FCFA

Government contract, all operators

Then CEMAC



F2G plan — to be validated with the team

NEXT STEPS

From review to pilot

1 **Team review**
Scope, pricing and roles

2 **Live lab demo**
Presidential, AMBER, ETWS

3 **MINPOSTEL / ART meeting**
National warning framework

4 **One-city operator pilot**
MTN or Orange

OUR ASK

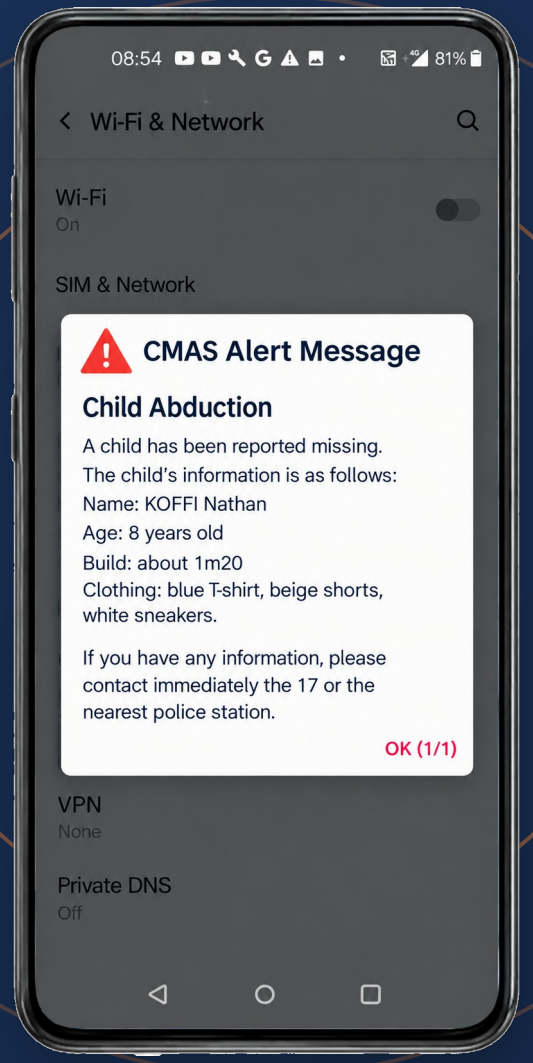
Your feedback, and one contact at MINPOSTEL, ART or an operator.

CMAS ON OPEN5GS + SRSRAN

Live demonstration

A real alert, broadcast from our Portable Lab to a test phone.

Presidential **AMBER** **ETWS**



APPENDIX

Technical appendix

Standards, architecture, call flow, hub

18 What is CMAS / PWS

19 International standards

20 Alert types and message IDs

21 3GPP architecture

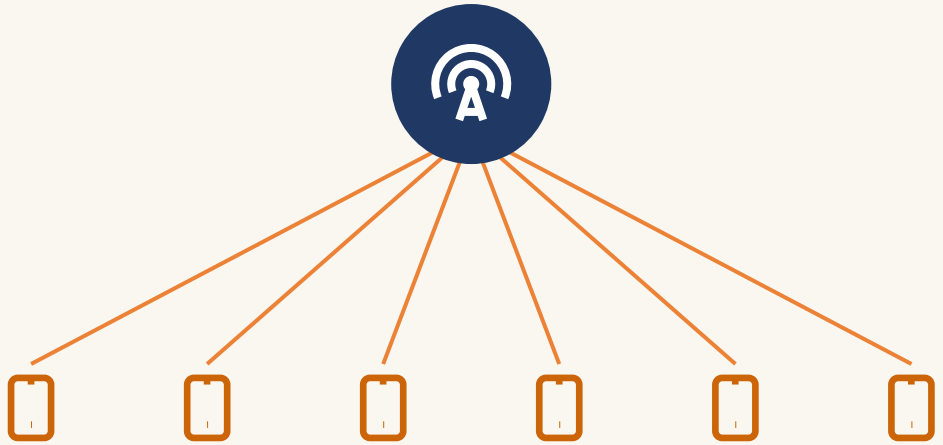
22 End-to-end call flow

23 The alert journey

24 The F2G CMAS Hub

Cell broadcast reaches an area; SMS, a list

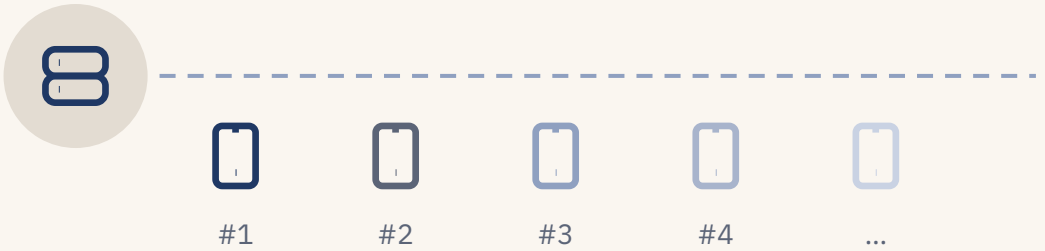
CELL BROADCAST (CMAS)



One message, every phone in the cells, at once

- No numbers, no SIM database
- Under 4 seconds, unlimited reach
- Full-screen popup and special sound

CLASSIC SMS



One message per number, one by one

- Needs every recipient number
- Minutes to hours at scale
- Easily missed among other texts

Cameroon would join WEA, EU-Alert, ETWS, KPAS



WEA

United States

Presidential, Imminent Threat and AMBER alerts



EU-Alert

European Union

Public warning under Directive (EU) 2018/1972



ETWS

Japan

Earthquake and tsunami warning



KPAS

South Korea

Nationwide cell-broadcast disaster alerts

All four run on 3GPP cell broadcast

3GPP TS 23.041 · one core technology, one platform for Cameroon

Six alert classes, fixed by 3GPP message IDs

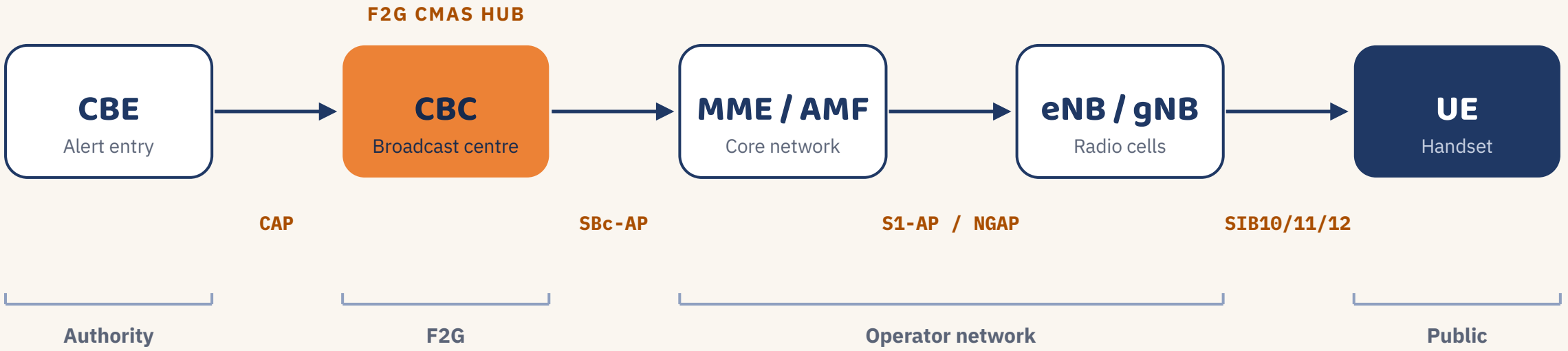
ALERT CLASS	MESSAGE ID	TYPICAL USE
Presidential	**4370**	National emergency — No opt-out
Extreme	**4371–4372**	Extreme threat, imminent or expected
Severe	**4373–4378**	Severe threat, e.g. weather or hazard
AMBER	**4379**	Child abduction
ETWS	**4352–4354**	Earthquake and tsunami
Test	**4380 / 4392**	Monthly or local test

Message identifiers per 3GPP TS 23.041; handsets already recognise them.

Five network elements from authority to handset

ALERT SOURCES

Government / MINDEF · Météo and Protection Civile · Ministry of Health, sending CAP messages



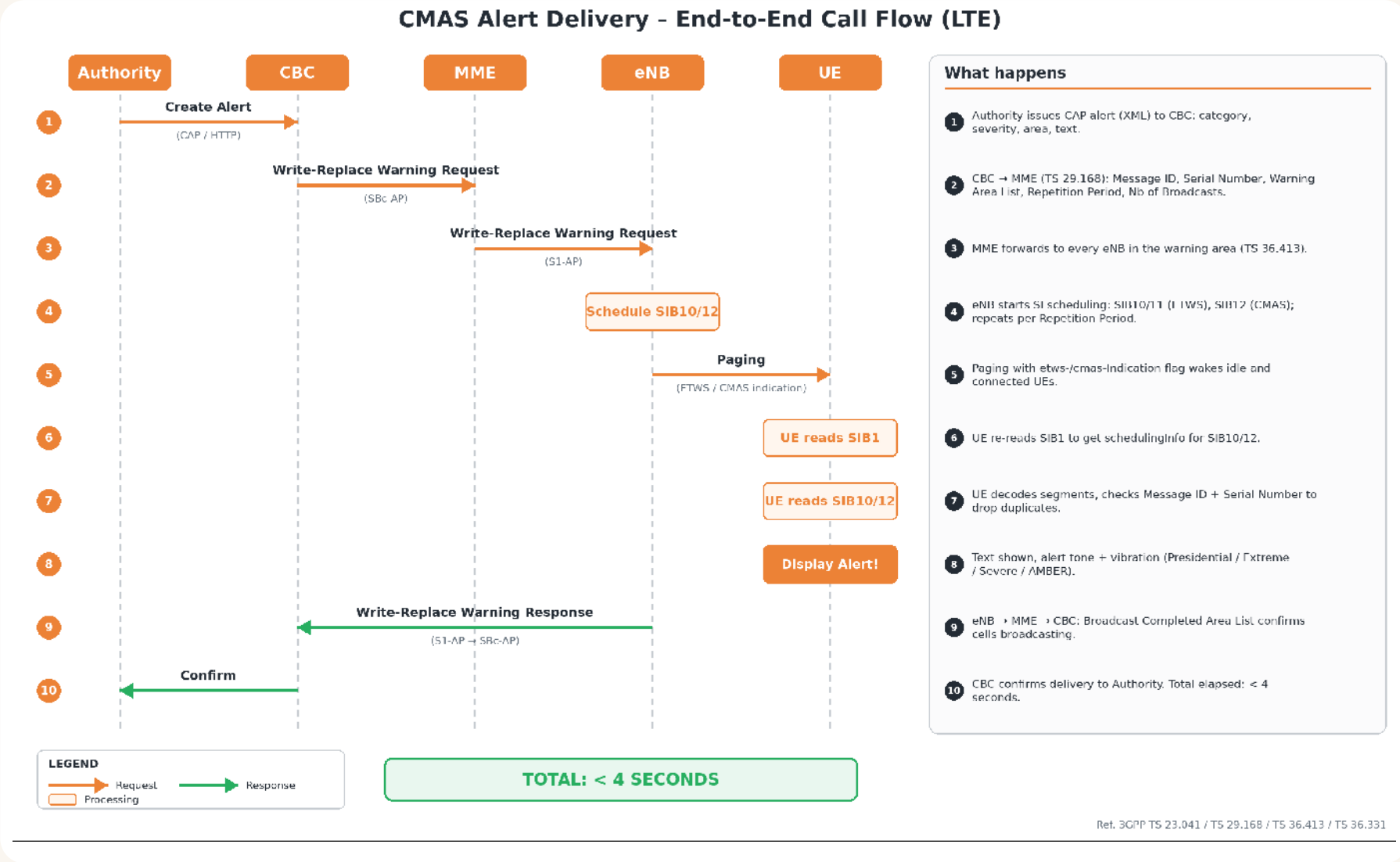
The CBC talks to each operator core over SBc-AP (3GPP TS 29.168). The core sends a Write-Replace Warning over S1-AP or NGAP; the cell broadcasts SIB10/11 (ETWS) and SIB12 (CMAS), and the phone shows the alert.

End-to-end call flow, under 4 s

1 Write-Replace Warning
CBC → MME → eNB

2 SIB12 and paging
eNB → every UE

3 Popup on screen
under 4 s in total



Four steps from decision to popup

1 Create

An authority writes and approves the alert and picks the target cells.

2 Broadcast

The CBC sends a Write-Replace Warning over SBc-AP to each MME / AMF.

3 Schedule

Each eNodeB schedules SIB12 and pages its cells with a CMAS indication.

4 Display

Every phone in the area shows a full-screen alert with a special sound.

< 4 s **end to end,** authority to handset

One hub between authorities and every operator

